

Assignment - 05

Section - A

1. Define partitioned clustering in data mining & give one eg. algorithm.

Ans. Partitioned clustering is a clustering approach where the dataset is divided into a fixed no. of non-overlapping clusters. Each data point belongs to exactly one cluster. The goal is to minimize intra-cluster distance & maximize inter-cluster distance. Eg: K-Means.

2. What is a dendrogram in hierarchical clustering?

Ans. A dendrogram is a tree-like diagram that displays the merging or splitting of clusters in hierarchical clustering. It shows how clusters are combined step by step and helps decide the no. of clusters by cutting the tree at a certain height.

3. Define Web content mining.

Ans. It is a process of extracting useful info. from the content of web pages, such as text, images, videos and structure data (tables, lists). It focuses on analyzing what is actually present on the page.

4. What is Web structure mining? Mention one typical application.

Ans. Web Structure Mining analyzes the hyperlink structure of the Web to identify relationships b/w pages. It uses graph theory to examine in-links, out-links & page connectivity.

Application: PageRank (used by Google for ranking pg.)

5. Write any two challenges in multimedia data mining.

Ans. i) Large size & high dimensionality of multimedia data
ii) Complexity in feature extraction from audio, video & images.

Section - B.

1. Explain density-based clustering. Describe concepts of core, border noise points & outline DBSCAN.

Ans. ~~Explain~~ density-based clustering groups data points that are closely packed together, while separating low-density regions are noise. It can detect clusters of arbitrary shape.

Key Concepts:

- Core Point $\rightarrow$ Has atleast M pts points within radius ϵ .

- Border Point $\rightarrow$ Has fewer than MinPts within Eps but lies in the neighborhood of a core point.
- Noise Point $\rightarrow$ Neither core nor border, considered as outlier.

Working of DBSCAN:

- Select an unvisited point.
- If core point $\rightarrow$ create a cluster.
- Expand the cluster by including density-reachable points.
- Mark remaining unclassified points as noise.

2. Explain basic process of text mining. Discuss text preprocessing, doc. representation & two tasks like classification & sentimental analysis.

Ans. Text mining extracts meaningful patterns from unstructured text.

- Text Preprocessing.
 - Tokenization
 - Stop-word removal
 - Stemming / Lemmatization
 - Removing punctuation, numbers
 - Lowercasing

ii) Document Representation

- Bag-of-words (BoW)
- TF-IDF
- Word embeddings (Word2Vec, GloVe).

iii) Text Mining Tasks

- Text classification: Assigning labels. eg: spam detection.
- Sentiment Analysis: Identifying emotion (pos, neg, neutral).

3. What are grid based clustering methods? Explain general approach & any one algorithm (STING or CLIQUE)

Ans: Grid-based methods quantize the data space into a finite no. of cells (grid structure) & perform clustering on the grid instead of individual data points. They offer high speed & work well for large datasets.

General Approach:

- Divide data space into grid cells.
- Compute statistical information for each cell.
- Identify dense cells.
- Merge adjacent dense cells to form clusters.

Eg: STING (Statistical Information Grid)

- Divides data into hierarchical grid cells.

- Stores statistical parameters (mean, count, etc) for each cell
- Uses top-down approach to identify dense regions.
- Fast & scalable but depends on granularity.

Section - C

1. Discuss Web mining in detail: Define Web content-mining, structure mining & usage mining with sources, methods & applications.

Ansⁿ Web mining is the process of discovering patterns from Web data, including content, structure and user behaviour.

a) Web Content Mining

→ Sources of Data

- Web pages (HTML text)
- Images, audio, video
- Metadata
- Blogs, news articles

→ Methods

- Text mining (tokenization, parsing)
- Multimedia mining
- Wrapper extraction

→ Applications

- Product recommendation
- Document classification.
- Search engine data extraction.

b) Web Structure Mining

→ Sources of Data

- Hyperlink connections b/w pages.

→ Methods

- Graph theory
- Link analysis (PageRank, HITS)

→ Applications

- Ranking web pages
- Finding authoritative hubs
- Community detection.

c) Web Usage Mining

→ Sources of Data

- Web server logs
- Browser logs
- Cookies
- User session histories.

→ Methods

- Session identification
- Path analysis
- Clustering user behaviour.

→ Applications

- Personalised recommendations.
- Website optimization
- Fraud detection.

2. Write detailed notes on spatial and multimedia data mining. Describe characteristics, techniques, applⁿ and challenges.

Ans. a) Spatial Data Mining: Analyzes spatial data that has geographic or location based relationships.

→ Characteristics

- Spatial autocorrelation
- Spatial heterogeneity
- Multi-dimensional structure.

→ Techniques

- Spatial clustering
- Spatial Classification

- Spatial association rules.

- Applications

- GIS mapping
- Urban planning
- Agriculture & Land use analysis
- Disaster management

- Challenges

- Huge data volume
- Complex spatial relationships
- Real-time processing requirements

b) Multimedia Data Mining

- Characteristics

- High dimensional
- Multiple datatypes (Image, video, audio)
- Temporal and semantic information.

- Techniques

- Feature extraction (color, texture, shape)
- Video segmentation
- Audio signal processing
- Deep learning (CNNs)

- Applications
 - Medical imaging
 - Surveillance
 - Remote sensing
 - Social media analytics.

- Challenges
 - Semantic gap
 - Storage and computation cost
 - Noise and variability

3. Elaborate the complete pipeline of text mining with a case study (sentiment analysis). Explain data collection, preprocessing, feature extraction, model building, evaluation & extensions.

Ans. a) Data Collection

- Social media posts (Twitter, Instagram).
- Online reviews.
- Blogs and comments.

b) Preprocessing

- Remove stop words
- Tokenize text
- Stemming / Lemmatization

- Handling emojis, hashtags.

c) Feature Extraction

- Bag of Words
- TF-IDF
- Word embeddings (Word2Vec, BERT)

d) Model Building

- Machine learning models (SVM, Naive Bayes)
- Deep learning (RNN, LSTM, Transformers)

e) Evaluation

- Accuracy
- Precision / Recall
- F2-Score
- Confusion matrix.

f) Extensions

- Aspect-based sentiment analysis
- Sarcasm detection
- Multilingual sentiment analysis
- Real-time sentiment dashboards.